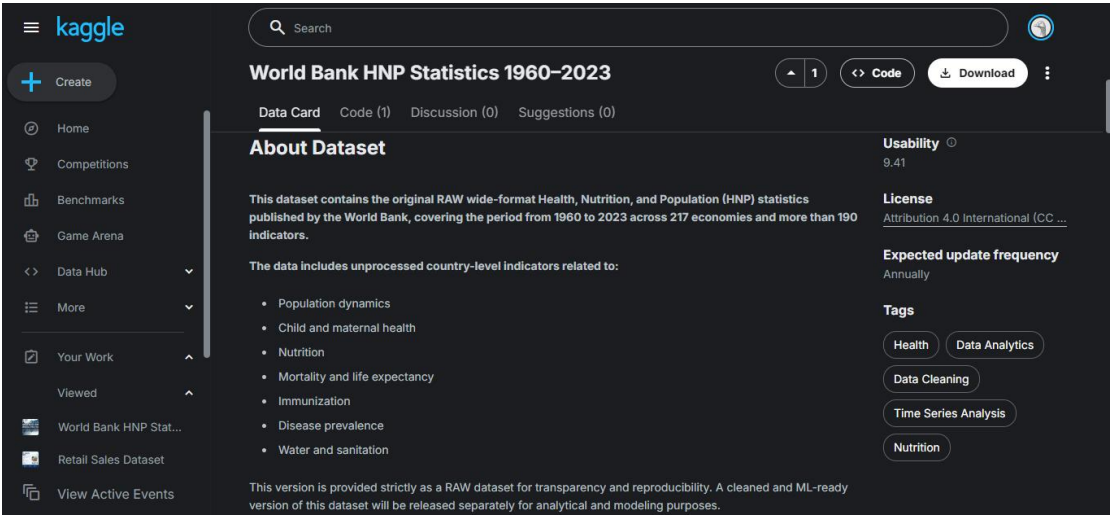


README.md

Verification

URL: [https://www.kaggle.com/datasets/indrajithsudusinghe/world-bank-hnp-statistics-19602023-raw-dataset/\[reference:8\]](https://www.kaggle.com/datasets/indrajithsudusinghe/world-bank-hnp-statistics-19602023-raw-dataset/[reference:8])

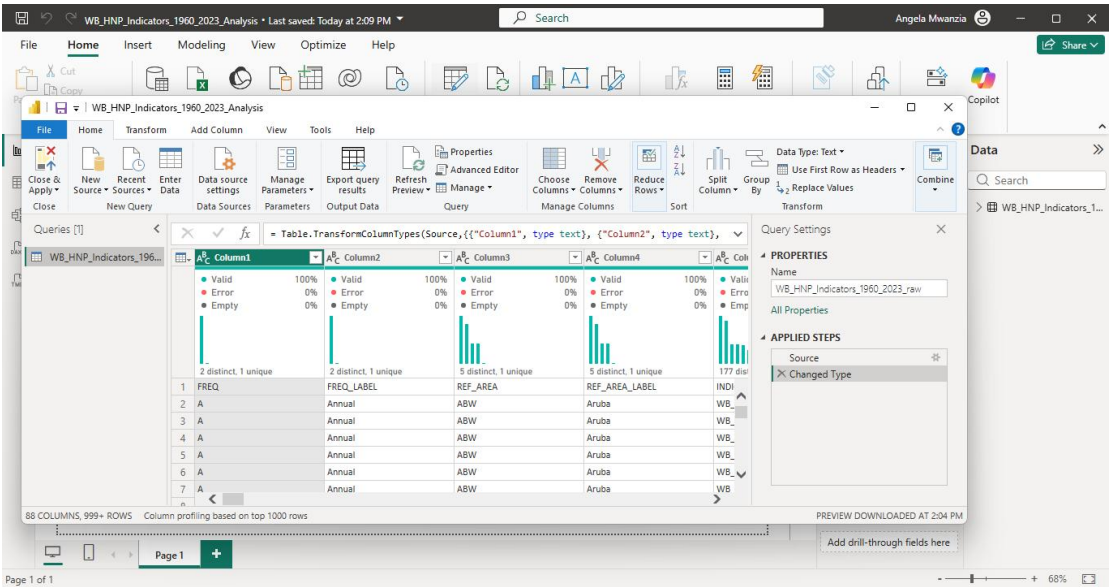


This version is provided strictly as a RAW dataset for transparency and reproducibility. A cleaned and ML-ready version of this dataset will be released separately for analytical and modeling purposes.

Source:
World Bank – Health, Nutrition and Population Statistics (HNP)
https://data360.worldbank.org/en/dataset/WB_HNP

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LOAD AND TRANSFORM DATASET



Question 1: Advanced Power Query Data Preparation — 60 Marks

Use Power Query to clean, transform, and prepare the dataset.

A. Basic Data Cleaning — 15 Marks

1. Rename unclear columns. (2 Marks)

Queries [1] = Table.RenameColumns(#"Changed Type1",{{"REF_AREA", "COUNTRY_CODE"}, {"REF_AREA_LABEL", "COUNTRY"}, {"INDICATOR", "INDICATOR_CODE"}, {"INDICATOR", "INDICATOR_CODE"}])

WB_HNP_Indicators_196...

	FREQ	FREQ_LABEL	COUNTRY_CODE	COUNTRY	INDICATOR_CODE	INDICATOR
1	A	Annual	ABW	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per cap
2	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
3	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
4	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
5	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
6	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
7	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
8	A	Annual	ABW	Aruba	WB_HNP_SE_ENRR	School enroll
9	A	Annual	ABW	Aruba	WB_HNP_SE_ENRR	School enroll
10	A	Annual	ABW	Aruba	WB_HNP_SE_ENRR	School enroll
11	A	Annual	ABW	Aruba	WB_HNP_SE_ENRR	School enroll
12	A	Annual	ABW	Aruba	WB_HNP_SE_ENRR	School enroll
13	A	Annual	ABW	Aruba	WB_HNP_SE_ENRR	School enroll
14	A	Annual	ABW	Aruba	WB_HNP_SE_ENRR	School enroll
15	A	Annual	ABW	Aruba	WB_HNP_SE_ENRR	School enroll
16	A	Annual	ABW	Aruba	WB_HNP_SE_NENR	School enroll

Query Settings

PROPERTIES

Name

WB_HNP_Indicators_1960_2023_raw

APPLIED STEPS

Source

Changed Type

Promoted Headers

Changed Type1

Renamed Columns

2. Change data types correctly. (2 Marks)

Queries [1] = Table.TransformColumnTypes(#"Renamed Columns",{{"FREQ", type text}})

WB_HNP_Indicators_196...

	FREQ	FREQ_LABEL	COUNTRY_CODE	COUNTRY	INDICATOR_CODE	INDICATOR
1	A	Annual	ABW	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per cap
2	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
3	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
4	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
5	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
6	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
7	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
8	A	Annual	ABW	Aruba	WB_HNP_SE_ENRR	School enroll

Query Settings

PROPERTIES

Name

WB_HNP_Indicators_1960_2023_raw

APPLIED STEPS

Source

Changed Type

Promoted Headers

Changed Type1

Renamed Columns

Changed Type2

3. Remove duplicate records. (2 Marks)

Queries [1] = Table.Distinct(#"Changed Type2")

WB_HNP_Indicators_196...

	FREQ	FREQ_LABEL	COUNTRY_CODE	COUNTRY	INDICATOR_CODE	INDICATOR
1	A	Annual	ABW	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per cap
2	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
3	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
4	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate
5	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LITR_ZS	Literacy rate

Query Settings

PROPERTIES

Name

WB_HNP_Indicators_1960_2023_raw

APPLIED STEPS

Source

Changed Type

Promoted Headers

Renamed Columns

Changed Type2

Removed Duplicates

4. Remove blank rows. (2 Marks)

Queries [1] < X ✓ fx = Table.SelectRows(#"Removed Duplicates", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"", null})))

WB_HNP_Indicators_196...

A ⁰	FREQ	A ⁰	FREQ_LABEL	A ⁰	COUNTRY_CODE	A ⁰	COUNTRY	A ⁰	INDICATOR_CODE	A ⁰	INDICATOR
1	A	Annual	ABW	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita					
2	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_UTR_ZS	Literacy rate					
3	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_UTR_ZS	Literacy rate					
4	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_UTR_ZS	Literacy rate					
5	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LI_FM_ZS	Literacy rate					
6	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LI_ZS	Literacy rate					
7	A	Annual	ABW	Aruba	WB_HNP_SE_ADT_LI_ZS	Literacy rate					

Query Settings

PROPERTIES

Name

WB_HNP_Indicators_1960_2023_raw

APPLIED STEPS

Source

Changed Type

Promoted Headers

Changed Type1

Renamed Columns

Changed Type2

Removed Duplicates

Removed Blank Rows

5. Trim and clean text columns. (2 Marks)

Queries [1] < X ✓ fx = Table.TransformColumns(#"Trimmed Text",{"(FREQ_LABEL", Text.Clean, type text), ("COUNTRY_CODE", Text.Clean, type text), ("COUNTRY", Text.Clean, type text), ("INDICATOR_CODE", Text.Clean, type text), ("INDICATOR", Text.Clean, type text), ("SEX", Text.Clean, type text), ("AGE", Text.Clean, type text), ("AGE_GROUP", Text.Clean, type text), ("URBANISATION", Text.Clean, type text), ("URBANISATION_LABEL", Text.Clean, type text), ("UNIT_MEASURE", Text.Clean, type text), ("UNIT_MEASURE_LABEL", Text.Clean, type text)})

WB_HNP_Indicators_196...

A ⁰	FREQ_LABEL	A ⁰	COUNTRY_CODE	A ⁰	COUNTRY	A ⁰	INDICATOR_CODE	A ⁰	INDICATOR	A ⁰	SEX
1	Annual	ABW	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method	T					
2	Annual	ABW	Aruba	WB_HNP_SE_ADT_UTR_ZS	Literacy rate, adult total (% of pop. 15+)	F					
3	Annual	ABW	Aruba	WB_HNP_SE_ADT_UTR_ZS	Literacy rate, adult total (% of pop. 15+)	M					
4	Annual	ABW	Aruba	WB_HNP_SE_ADT_UTR_ZS	Literacy rate, adult total (% of pop. 15+)	T					
5	Annual	ABW	Aruba	WB_HNP_SE_ADT_LI_FM_ZS	Literacy rate, youth (ages 15-24)	T					
6	Annual	ABW	Aruba	WB_HNP_SE_ADT_LI_ZS	Literacy rate, youth (% of pop. 15+)	M					
7	Annual	ABW	Aruba	WB_HNP_SE_ADT_LI_ZS	Literacy rate, youth (% of pop. 15+)	T					

Query Settings

PROPERTIES

Name

WB_HNP_Indicators_1960_2023_raw

APPLIED STEPS

Source

Changed Type

Promoted Headers

Changed Type1

Renamed Columns

Changed Type2

Removed Duplicates

Removed Blank Rows

Trimmed Text

Cleaned Text

6. Replace inconsistent values. (2 Marks)

Replace Values

Replace one value with another in the selected columns.

Value To Find

Total

Replace With

Both

Advanced options

OK

Cancel

Queries [1] < X ✓ fx = Table.ReplaceValue(#"Cleaned Text", "Total", "Both", Replacer.ReplaceText, {"SEX_LABEL"})

WB_HNP_Indicators_196...

A ⁰	SEX_LABEL	A ⁰	AGE	A ⁰	AGE_GROUP	A ⁰	URBANISATION	A ⁰	URBANISATION_LABEL	A ⁰
1	Both	T	All age ranges or no breakdown by age	T	Total					
2	Female	T	All age ranges or no breakdown by age	T	Total					
3	Male	T	All age ranges or no breakdown by age	T	Total					
4	Both	T	All age ranges or no breakdown by age	T	Total					
5	Both	Y15T24	15 to 24 years old	T	Total					
6	Male	Y15T24	15 to 24 years old	T	Total					
7	Both	Y15T24	15 to 24 years old	T	Total					

Query Settings

PROPERTIES

Name

WB_HNP_Indicators_1960_2023_raw

APPLIED STEPS

Source

Changed Type

Promoted Headers

Changed Type1

Renamed Columns

Changed Type2

Removed Duplicates

Removed Blank Rows

Trimmed Text

Cleaned Text

Replaced Value

7. Remove unnecessary columns. (3 Marks)

Queries [1] WB_HNP_Indicators_196...

Table.RemoveColumns("#Replaced Value",{"DATABASE_ID"})

Database ID Label	Unit Mult	Unit Mult Label	Obs Status	Obs Status Label
1 distinct, 0 unique	1 distinct, 0 unique	1 distinct, 0 unique	1 distinct, 0 unique	1 distinct, 0 unique
1 Health Nutrition and Population Statistics	0	Units	A	Normal value
2 Health Nutrition and Population Statistics	0	Units	A	Normal value
3 Health Nutrition and Population Statistics	0	Units	A	Normal value
4 Health Nutrition and Population Statistics	0	Units	A	Normal value
5 Health Nutrition and Population Statistics	0	Units	A	Normal value
6 Health Nutrition and Population Statistics	0	Units	A	Normal value
7 Health Nutrition and Population Statistics	0	Units	A	Normal value
8 Health Nutrition and Population Statistics	0	Units	A	Normal value

Query Settings

PROPERTIES

Name

WB_HNP_Indicators_1960_2023_raw

All Properties

APPLIED STEPS

Source

Changed Type

Promoted Headers

Changed Type1

Renamed Columns

Changed Type2

Removed Duplicates

Removed Blank Rows

Trimmed Text

Cleaned Text

Replaced Value

Removed Columns

B. Intermediate Transformations — 20 Marks

1. Split one column into multiple columns. (2 Marks)

Split Column by Delimiter

Specify the delimiter used to split the text column.

Select or enter delimiter

--Custom--

and

Split at

☐ Left-most delimiter

☐ Right-most delimiter

☒ Each occurrence of the delimiter

Advanced options

Quote Character

"

☐ Split using special characters

Insert special character

OK Cancel

Table.TransformColumnTypes("#Split Column by Delimiter",{{"DATABASE_ID_LABEL.1", type text}, {"DATABASE_ID_LABEL.2", type text}})

2. Merge two or more columns. (2 Marks)

Merge Columns

Choose how to merge the selected columns.

Separator

Colon

New column name (optional)

COUNTRY AND CODE

OK

Cancel

```
= Table.CombineColumns("#Changed Type3",{ "COUNTRY_CODE", "COUNTRY"},Combiner.CombineTextByDelimiter(":",  
QuoteStyle.None),"COUNTRY AND CODE")
```

3. Create a custom column. (3 Marks)

Custom Column

Add a column that is computed from the other columns.

New column name

TOTAL_FIRST_10_YRS

Custom column formula ⓘ

```
= [1960]+[1961]+[1962]+[1963]+[1964]+[1965]+[1966]+[1967]+  
[1968]+[1969]+[1970]
```

Available columns

1964
1965
1966
1967
1968
1969
1970

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

Custom Column

×

Add a column that is computed from the other columns.

New column name

TOTAL_LAST_10_YEARS

Custom column formula ⓘ

= [2013]+[2014]+[2015]+[2016]+[2017]+[2018]+[2019]+[2020]+[2021]+[2022]+[2023]

Available columns

2017

2018

2019

2020

2021

2022

2023

TOTAL_FIRST_10_YRS

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

ABC 123	TOTAL_FIRST_10_YRS	ABC 123	TOTAL_LAST_10_YEARS
Valid	33%	Valid	31%
Error	0%	Error	0%
Empty	67%	Empty	69%
	149.1049034		77.52806681
	118.069911		76.4866355
	601.147715		752.6588575
	89.11568505		69.94257423
	74.46875338		66.51254878
	65.2240223		67.71115563
	56.42059214		69.13267177
	49.80644333		73.82842885
	45.61651984		80.70537244
	43.04659925		88.20144506
	35.68123528		86.20670268
	162.6033418		72.20239397

4. Create a conditional column. (3 Marks)

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

HEALTH_CATEGORY

Column Name

Operator

Value

Output

If

INDICATOR

contains

ABC 123 Mortality

Then

ABC 123 Mortality

Else If

INDICATOR

contains

ABC 123 Immunization

Then

ABC 123 Immunization

Else If

INDICATOR

contains

ABC 123 Nutrition

Then

ABC 123 Nutrition

...

Add Clause

Else

ABC 123

Other

OK

Cancel

Queries [1]

WB_HNP_Indicators_196...

Table.AddColumn(#"Added Custom1", "HEALTH_CATEGORY", each if Text.Contains([INDICATOR], "Mortality") then "Mortality" else if Text.Contains([INDICATOR], "Immunization") then "Immunization" else if Text.Contains([INDICATOR], "Nutrition") then "Nutrition" else "Other")

1.2 2022

1.2 2023

ABC 123 TOTAL_FIRST_10_YRS

ABC 123 TOTAL_LAST_10_YEARS

ABC 123 HEALTH_CATEGORY

69% Valid 0% Error 31% Empty

64% Valid 0% Error 36% Empty

31% Valid 0% Error 69% Empty

33% Valid 0% Error 67% Empty

31% Valid 0% Error 69% Empty

100% Valid 0% Error 0% Empty

ue

62k distinct, 612 unique

315 distinct, 314 unique

933 null null null null null Other

934 null null null null null Other

935 10089179 70.51367706 null null null Immunization

936 10089179 70.51367706 null null null Immunization

937 40890537 79.35454431 null null null Immunization

938 10026679 null null null null null Immunization

939 49931435 39.33745312 null null null Immunization

940 64636102 65.78037582 null null null Immunization

941 51633847 68.20355585 null null null Immunization

942 null null null null null Other

943 null null null null null Other

944 null null null null null Other

Query Settings

PROPERTIES

Name

WB_HNP_Indicators_1960_2023_raw

All Properties

APPLIED STEPS

Source

Changed Type

Promoted Headers

Changed Type1

Renamed Columns

Changed Type2

Removed Duplicates

Removed Blank Rows

Trimmed Text

Cleaned Text

Replaced Value

Removed Columns

Split Column by Delimiter

Changed Type3

Merged Columns

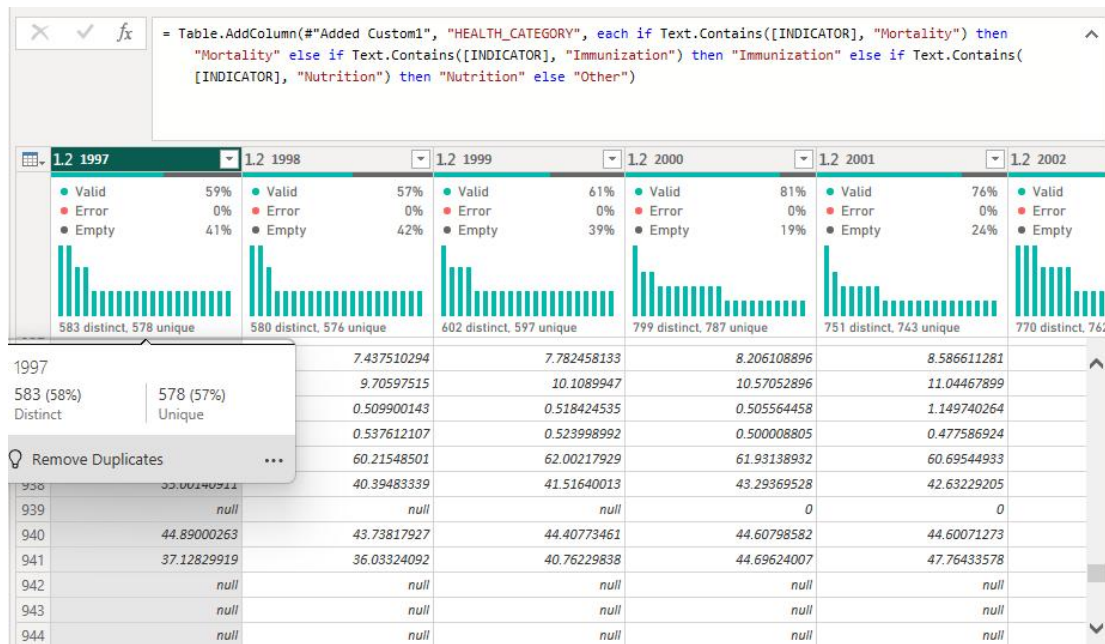
Added Custom

Added Custom1

Added Conditional Column

5. Extract year, month, quarter, and day from a date column. (4 Marks)

Data is pivoted by year



6. Filter rows using at least two conditions. (2 Marks)

Filter Rows

Apply one or more filter conditions to the rows in this table.

☒ Basic ☐ Advanced

Keep rows where 'INDICATOR'

contains Population

☒ And ☐ Or

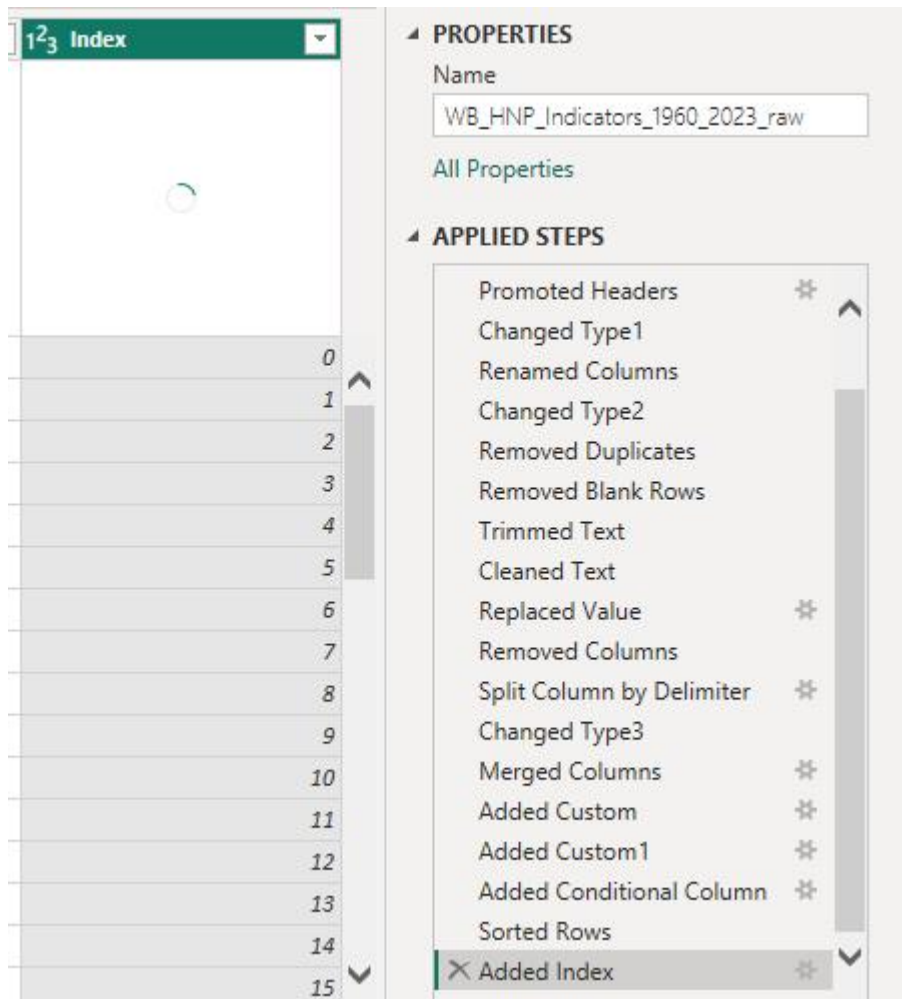
contains Literacy

OK Cancel

7. Sort data meaningfully. (2 Marks)

`= Table.Sort("#Added Conditional Column",{"COUNTRY AND CODE", Order.Ascending})`

8. Add an index column. (2 Marks)



C. Advanced Power Query Tasks — 25 Marks

Complete any five of the following tasks.

Each task carries 5 marks.

1. Merge queries using a common key.
2. Append two related datasets.
3. Pivot a column.
4. Unpivot columns.

Query Settings

Table.Unpivot(#Added Index, {"FREQ", "1960", "1961", "1962", "1963", "1964", "1965", "1966", "1967", "1968",

WB_HNP_Indicators_1960_2023_raw

APPLIED STEPS

- Changed Type1
- Renamed Columns
- Changed Type2
- Removed Duplicates
- Removed Blank Rows
- Trimmed Text
- Cleaned Text
- Replaced Value
- Removed Columns
- Split Column by Delimiter
- Changed Type3
- Merged Columns
- Added Custom
- Added Custom1
- Added Conditional Column
- Sorted Rows
- Added Index
- Unpivoted Only Selected C...

10_YRS	TOTAL_LAST_10_YEARS	HEALTH_CATEGORY	Index	Attribute	Value
1	null	null	Other	0	FREQ
2	null	null	Other	0	1988
3	null	null	Other	0	1989
4	null	null	Other	0	1990
5	null	null	Other	0	1991
6	null	null	Other	0	1992
7	null	null	Other	0	1993
8	null	null	Other	0	1994
9	null	null	Other	0	1995
10	null	null	Other	0	1996
11	null	null	Other	0	1997
12	null	null	Other	0	1998
13	null	null	Other	0	1999
14	null	null	Other	0	2000
15	null	null	Other	0	2001
16	null	null	Other	0	2002

nn profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 8:30 PM

5. Create a parameter and use it to filter data.

6. Create a date table in Power Query.

Queries [2]

WB_HNP_Indicators_1960_2023_raw

DATE_TABLE

Table.AddColumn(#"Added Custom3", "DAY_OF_WEEK", each Date.DayOfWeek([DATE]))

DATE	YEAR	MONTH	QUARTER	MONTH_NAME	DAY_OF_WEEK
1	1/1/1960	1960	1	January	
2	1/2/1960	1960	1	January	
3	1/3/1960	1960	1	January	
4	1/4/1960	1960	1	January	
5	1/5/1960	1960	1	January	
6	1/6/1960	1960	1	January	
7	1/7/1960	1960	1	January	
8	1/8/1960	1960	1	January	
9	1/9/1960	1960	1	January	

365 distinct, 365 unique

APPLIED STEPS

- Source
- Converted to Table
- Renamed Columns
- Changed Type
- Added Custom
- Added Custom1
- Added Custom2
- Added Custom3
- Added Custom4

7. Use Group By with multiple aggregations.

Group By

Specify the columns to group by and one or more outputs.

Basic

Advanced

COUNTRY AND CODE

Add grouping

New column name

FIRST_10_YRS_TOTAL_BY_CON

LAST_10_YRS_TOTAL_BY_CON1

Add aggregation

Operation

Sum

Sum

Column

TOTAL_FIRST_10_YRS

TOTAL_LAST_10_YEARS

OK

Cancel

Queries [2] WB_HNP_Indicators_196... DATE_TABLE

Table.Group(#Unpivoted Only Selected Columns", {"COUNTRY AND CODE"}, {"FIRST_10_YRS_TOTAL_BY_CONTRY", each

	COUNTRY AND CODE	FIRST 10 YRS TOTAL BY CON...	LAST 10 YRS TOTAL BY CON...
1	ABW:Aruba	249543127.5	452185499.1
2	AFE: Africa Eastern and South...	6.45261E+11	2.89267E+12
3	AFG: Afghanistan	41240798860	1.61315E+11
4	APW: Africa Western and Cen...	4.66351E+11	1.96042E+12
5	AGO: Angola	24423467617	1.3921E+11
6	ALB: Albania	8016366137	12690645387
7	AND: Andorra	61499610.69	323159236.4
8	ARB: Arab World	4.59024E+11	1.90151E+12
9	ARE: United Arab Emirates	812842955.5	41601092872
10	ARG: Argentina	94675196828	1.98009E+11
11	ARM: Armenia	9564896098	12681225256
12	ASM: American Samoa	101292709.2	207836057.5
13	ATG: Antigua and Barbuda	257864508.8	401168794.7
14	AUS: Australia	48889556925	1.11356E+11
15	AUT: Austria	31167741840	39456967347
16	AZE: Azerbaijan	19546916871	44212051903
17	BDI: Burundi	13369298567	51630304264

Query Settings

PROPERTIES

Name

WB_HNP_Indicators_1960_2023_raw

All Properties

APPLIED STEPS

Changed Type1

Renamed Columns

Changed Type2

Removed Duplicates

Removed Blank Rows

Trimmed Text

Cleaned Text

Replaced Value

Removed Columns

Split Column by Delimiter

Changed Type3

Merged Columns

Added Custom

Added Custom1

Added Conditional Column

Sorted Rows

Added Index

Unpivoted Only Selected C...

Grouped Rows

8. Extract text before or after a delimiter.

Text Before Delimiter

Enter the delimiter that marks the end of what you would like to extract.

Delimiter

comma

Advanced options

OK Cancel

Table.TransformColumns(#Unpivoted Only Selected Columns", {"INDICATOR", each Text.BeforeDelimiter(_, "comma"), type text})

	COUNTRY AND CODE	INDICATOR_CODE	INDICATOR	SEX	SEX_LABEL
1	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both
2	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both
3	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both
4	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both
5	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both
6	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both
7	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both
8	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both
9	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both
10	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both
11	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both
12	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both
13	Aruba	WB_HNP_NY_GNP_PCAP_CD	GNI per capita, Atlas method (current US\$)	_T	Both

Query Settings

PROPERTIES

Name

WB_HNP_Indicators_1960_2023_raw

All Properties

APPLIED STEPS

Changed Type2

Removed Duplicates

Removed Blank Rows

Trimmed Text

Cleaned Text

Replaced Value

Removed Columns

Split Column by Delimiter

Changed Type3

Merged Columns

Added Custom

Added Custom1

Added Conditional Column

Sorted Rows

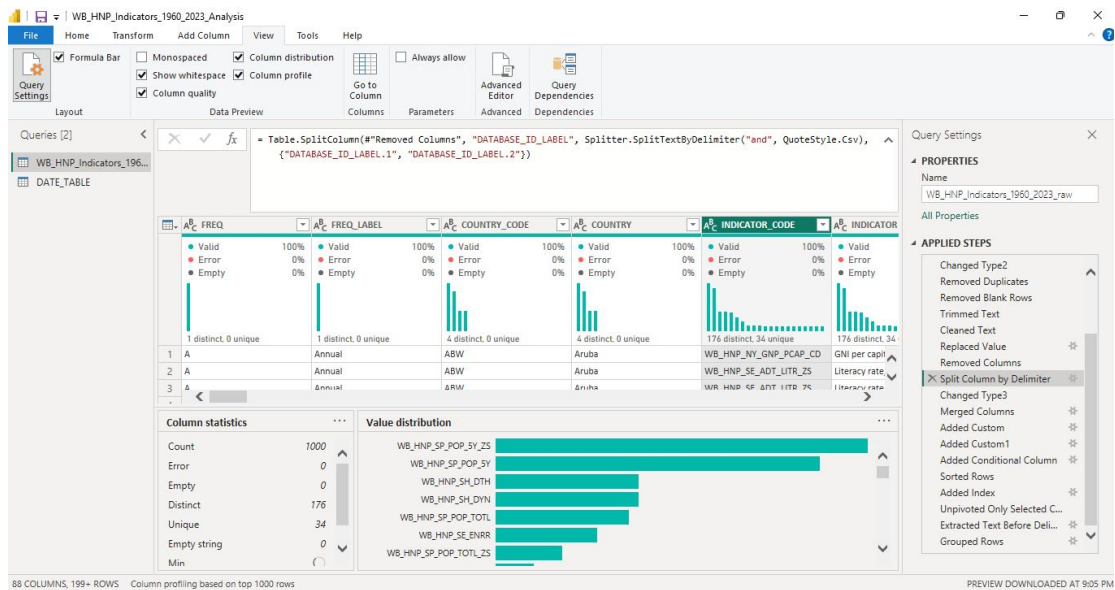
Added Index

Unpivoted Only Selected C...

Extracted Text Before Deli...

Grouped Rows

9. Remove or handle errors in a column.
10. Create business categories using nested conditional logic.
11. Create a summarized query using Reference Query.
12. Use Column Profiling to identify data quality issues.



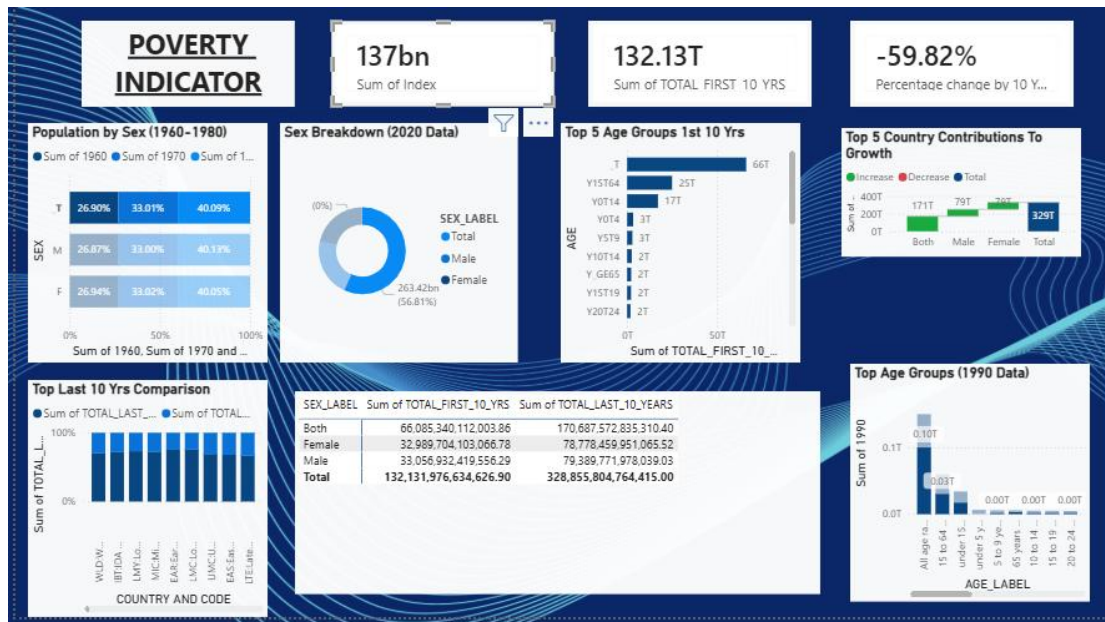
Power Query Evidence Required

Students must include screenshots of:

- Raw imported dataset
- Power Query Editor
- Applied Steps pane
- Column profiling results
- Final cleaned dataset
- Any merge, append, pivot, unpivot, or parameter task attempted

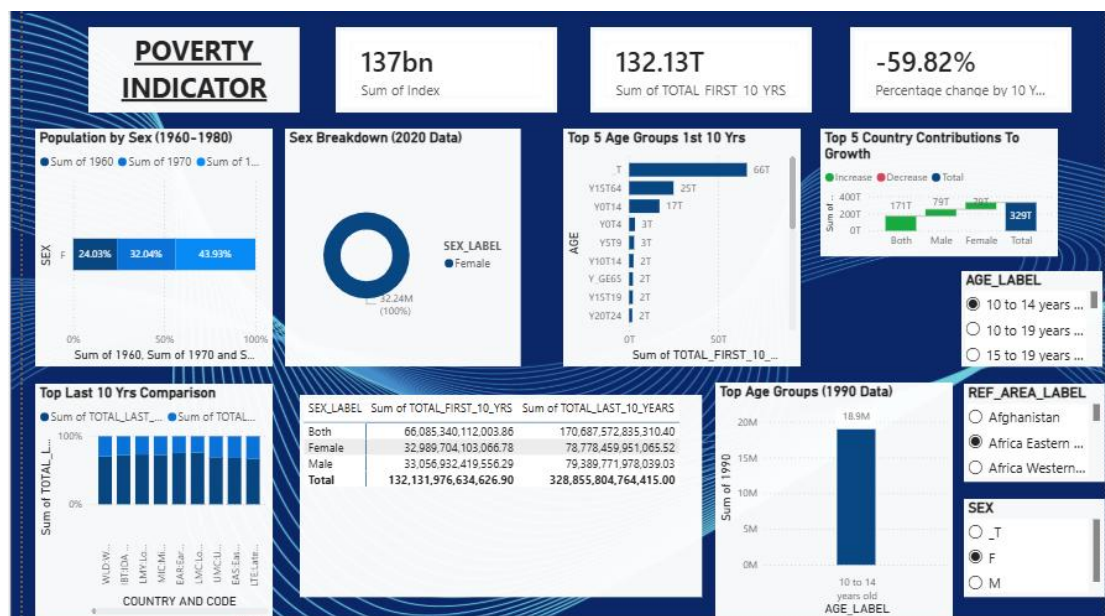
Question 2: Power BI Dashboard Development — 30 Marks

1. Create a professional Power BI dashboard using the cleaned dataset.
2. Required Visuals
3. Minimum 3 KPI cards. (3 Marks)
4. Bar chart. (3 Marks)
5. Column chart. (3 Marks)
6. Line chart showing a trend. (3 Marks)
7. Pie or donut chart. (3 Marks)
8. Table visual. (3 Marks)
9. Matrix visual. (3 Marks)
10. Map or filled map, where applicable. (3 Marks)
11. Any two additional meaningful visuals. (4 Marks)
12. Professional layout, titles, formatting, and readability. (2 Marks)



Question 3: Dashboard Interactivity and Insights — 10 Marks

1. Add at least three slicers. (3 Marks)
2. Enable drill-down on one visual. (2 Marks)
3. Demonstrate cross-filtering between visuals. (2 Marks)
4. Write three business insights based on the dashboard. (3 Marks)



Insight 1: Severe economic contraction in the measured indicator

The dashboard shows a drastic collapse, with the total for the first 10 years at 132.13T but the current Sum of Index dropping to just 137bn, representing a -59.82% negative change. This signals a massive structural decline in economic welfare or asset value, urging immediate macroeconomic intervention to halt the bleeding.

Insight 2: Significant demographic shift toward female prevalence

Over the 20-year period from 1960 to 1980, the female share of the population metric

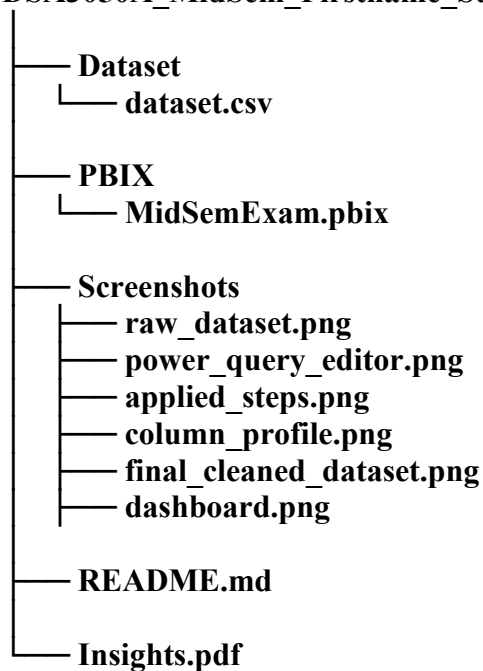
skyrocketed from 24.03% to 43.93% (nearly doubling). Combined with the 2020 Sex Breakdown, this indicates that women are becoming an increasingly dominant cohort in this dataset. Any recovery strategy must prioritize female-targeted financial inclusion and support programs to reverse this trend.

Insight 3: The working-age population bears the heaviest burden

The "Top 5 Age Groups" visual is overwhelmingly dominated by the Y15T64 (working-age, 15-64) cohort, eclipsing all other age bands. This implies that employment instability and labor market pressures are the primary drivers of the poverty indicator, meaning job creation and wage protection must be the central focus of any corrective business or governmental policy.

GitHub Repository Structure

DSA3050A_MidSem_Firstname_StudentID



README.md Requirements

The README file must contain:

Student name and student ID

Dataset source

Number of rows and columns

Business problem being analyzed

List of all Power Query transformations performed

List of visuals created

Three business insights from the dashboard